

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Department of Multidisciplinary
Engineering

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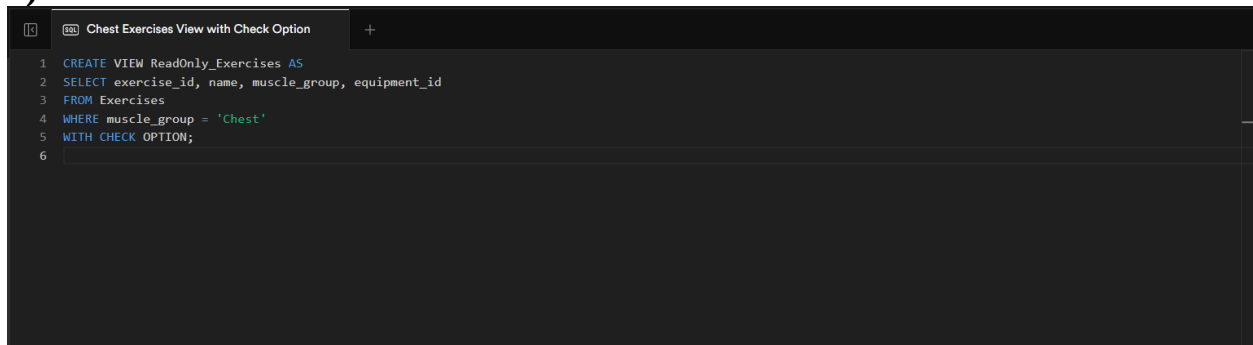
Title:

1. Create single table read-only view
2. Single table updatable view
3. Multitable view
4. View with aggregate function
5. Insert, delete and update operations through view
6. Create or replace view

Description: In this assignment, we implemented Views on the Workout Planner database using Exercises and Available_Equipments tables. We created read-only and updatable views, a multi-table join view, and a view with aggregate functions. We also performed insert, update, and delete operations through views and used CREATE OR REPLACE VIEW to modify existing ones. This showed how views simplify queries, enhance security, and present data clearly.

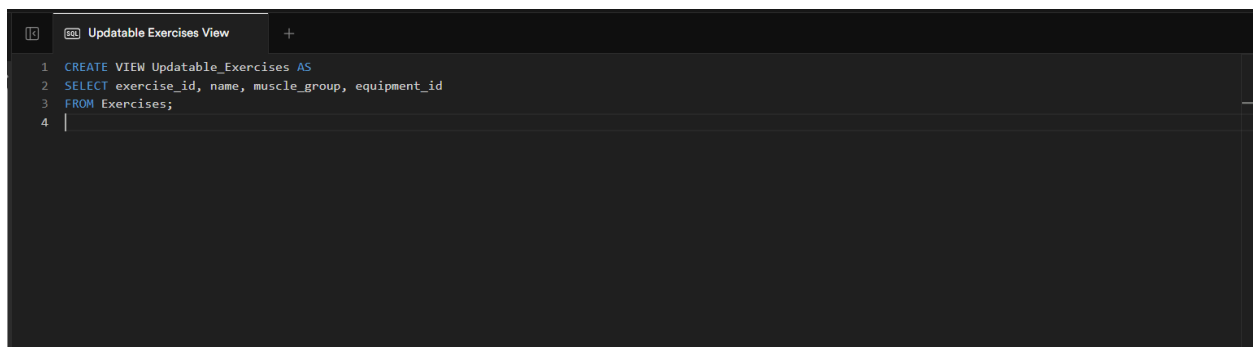
Screenshots/Output:

1)

A screenshot of a SQL editor window titled "Chest Exercises View with Check Option". The code defines a read-only view named "ReadOnly_Exercises" by selecting columns from the "Exercises" table where the muscle_group is 'Chest'.

```
1 CREATE VIEW ReadOnly_Exercises AS
2 SELECT exercise_id, name, muscle_group, equipment_id
3 FROM Exercises
4 WHERE muscle_group = 'Chest'
5 WITH CHECK OPTION;
```

2)

A screenshot of a SQL editor window titled "Updatable Exercises View". The code defines an updatable view named "Updatable_Exercises" by selecting columns from the "Exercises" table.

```
1 CREATE VIEW Updatable_Exercises AS
2 SELECT exercise_id, name, muscle_group, equipment_id
3 FROM Exercises;
```

3)

```
Exercise Equipment Mapping View +
1 CREATE VIEW Exercise_Equipment_View AS
2 SELECT e.exercise_id, e.name, e.muscle_group,
3        eq.equipment_name
4 FROM Exercises e
5 JOIN Available_Equipments eq
6     ON e.equipment_id = eq.equipment_id;
7
```

4)

```
Exercise Management Operations +
1 -- Insert a new exercise
2 INSERT INTO Updatable_Exercises (name, muscle_group, equipment_id)
3 VALUES ('Push Up', 'Chest', 1);
4
5 -- Update an exercise
6 UPDATE Updatable_Exercises
7 SET muscle_group = 'Shoulders'
8 WHERE exercise_id = 2;
9
10 -- Delete an exercise
11 DELETE FROM Updatable_Exercises
12 WHERE exercise_id = 3;
13
```

5)

```
Exercise Count By Muscle Group +
1 CREATE VIEW Exercise_Count_By_Muscle AS
2 SELECT muscle_group, COUNT(*) AS total_exercises
3 FROM Exercises
4 GROUP BY muscle_group;
5
```

6)

```
Exercise Equipment Mapping View +
1 CREATE OR REPLACE VIEW Exercise_Equipment_View AS
2 SELECT e.exercise_id, -- keep PK for safety
3        e.name,
4        e.muscle_group, -- extra column for detail
5        eq.equipment_name
6 FROM Exercises e
7 LEFT JOIN Available_Equipments eq
8     ON e.equipment_id = eq.equipment_id;
9
```

Outputs:

readonly_exercises

Filter Sort Security Definer view Role postgres API Docs

	exercise_id int4	name text	muscle_group text	equipment... i...
☐	9	Chest Fly	Chest	4
☐	11	Push Up	Chest	1

Page 1 of 1 100 rows 1.0 record (estimated) Refresh Data Definition

updatable_exercises

Filter Sort Security Definer view Role postgres API Docs

	exercise_id int4	name text	muscle_group text	equipment... i...
☐	1	Bench Press	Upper Chest	1
☐	2	Squat	Shoulders	2
☐	4	Bicep Curl	Biceps	4
☐	5	Tricep Pushdown	Triceps	3
☐	6	Shoulder Press	Shoulders	4
☐	7	Lat Pulldown	Back	3
☐	8	Leg Press	Legs	2
☐	9	Chest Fly	Chest	4
☐	11	Push Up	Chest	1

Page 1 of 1 100 rows 9.0 records (estimated) Refresh Data Definition

exercise_equipment_view

Filter

Sort

Security Definer view

Role: postgres

API Docs

☐	exercise_id int4	name text	muscle_group text	equipment_name text
☐	1	Bench Press	Upper Chest	Barbell
☐	2	Squat	Shoulders	Squat Rack
☐	4	Bicep Curl	Biceps	Adjustable Dumbbells
☐	5	Tricep Pushdown	Triceps	Cable Machine
☐	6	Shoulder Press	Shoulders	Adjustable Dumbbells
☒	7	Lat Pulldown	Back	Cable Machine
☐	8	Leg Press	Legs	Squat Rack
☐	9	Chest Fly	Chest	Adjustable Dumbbells
☐	11	Push Up	Chest	Barbell

Page 1 of 1

100 rows

9.0 records (estimated)

Refresh

Data

Definition

exercise_count_by_muscle

Filter

Sort

Security Definer view

Role: postgres

API Docs

☐	muscle_group text	total_exercises int8
☐	Back	1
☐	Biceps	1
☐	Chest	2
☐	Legs	1
☐	Shoulders	2
☐	Triceps	1
☐	Upper Chest	1

Page 1 of 1

100 rows

9.0 records (estimated)

Refresh

Data

Definition

* Views:-

In RDBMS, a view is a virtual table created from the result of a query. Unlike physical tables, views do not store data themselves; instead, they provide a way to look at the data stored in one or more underlying tables. Views are especially useful for simplifying complex queries, enforcing security & presenting only the required data to the user. In the context of our Workout Planner project, views are created on top of Exercises & Available Equipments tables to provide meaningful insights for users & administrators.

* Types of views:-

There are several types of views supported in SQL, each serving different purposes:

- 1) Simple views (single-table views):
These are created from a single table & usually display a subset of its columns.
- 2) Updatable views:
These allow users to perform INSERT, UPDATE & DELETE operations through view, provided they follow certain restrictions.
- 3) Complex views (Multi-table views):
These are created using joins across multiple tables. They combine related data, which makes them useful for reporting & analysis.

4) **Aggregate views:**
Created using aggregate functions like COUNT, SUM, AVG, MIN & MAX. They provide summarized information instead of detailed rows.

5) **Materialized views:**
Unlike normal views, materialized views store the query result physically. They are used for performance improvement but require refresh mechanisms.

* Advantages:-

- 1) **Data security:** views can restrict access to sensitive columns while exposing only necessary data to user.
- 2) **Simplification:** They hide query complexity by encapsulating joins & filters.
- 3) **Reusability:** once defined, view can be reused in multiple queries.
- 4) **Data aggregation:** views can summarize data & present ~~an~~ analytical reports directly.